Portfolio Credit Default Swaps

Seminar Module

conducted by

ACF Consultants Ltd





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Portfolio Credit Default Swaps

- A **single name** CDS involves just one reference entity or obligation
- A **portfolio** CDS involves two or more reference entities or obligations

There are two broad forms of portfolio CDS:

- Basket CDS
  - CDS is written on a portfolio of two or more reference names
  - If a credit event occurs:
    - Protection payment is made
    - Notional principal is reduced
    - CDS continues in existence…
- First-to-Default CDS (and similar structures)
  - CDS is written on a portfolio of two or more reference names
  - Notional principal of each reference entity is the same
  - If a credit event occurs:
    - Protection payment is made
    - CDS terminates

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How Indices Compare with Basket CDS

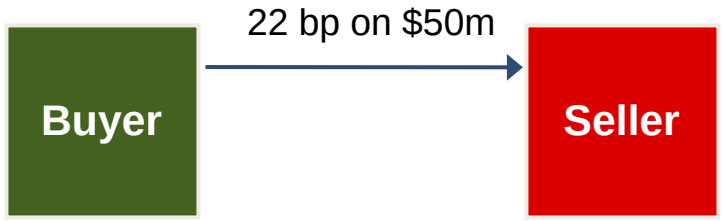
Indices	Basket CDS
Standardised basket of names	Basket can be tailored to suit counterparties' needs
Large basket of names (typically 30-125)	Small basket of names (5 or so)

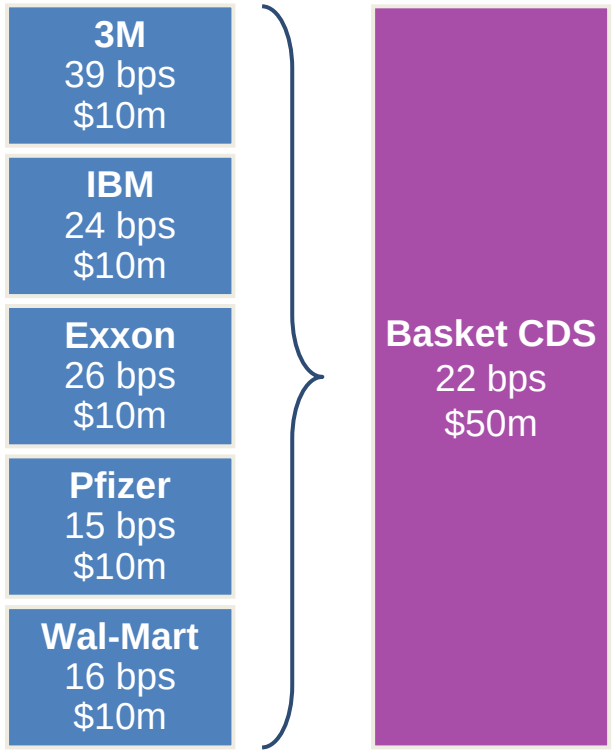
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Basket CDS Example

- Basket CDS on notional portfolio of $50m
- Comprises 5 names of $10m notional each
- Premium for basket less than average of individual CDS premiums:
  - Total premium for 5 separate CDS: 120 bp
  - Average premium per entity: 24 bp
  - Premium for basket CDS: 22 bp
- Premium paid each year:

22bp × $50m = $110,000 p.a.





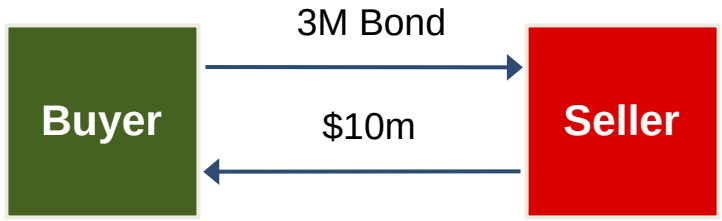
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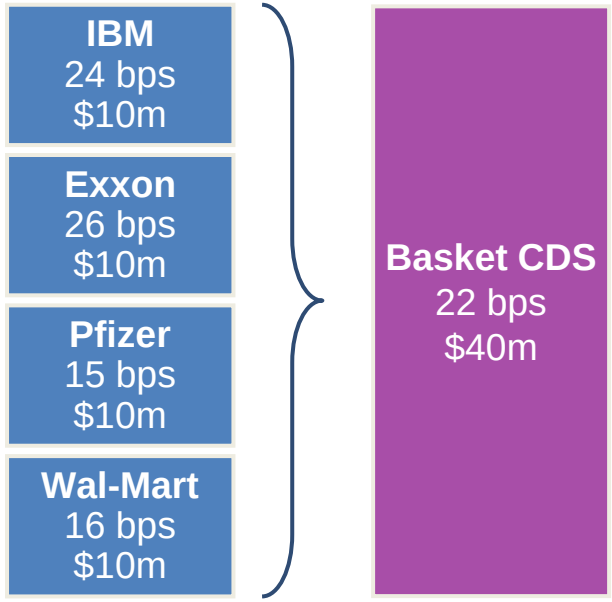
Basket CDS Example – After Default

Suppose 3M experiences a credit event…

- Basket CDS would make payment of $10m less recovery value
- Principal on Basket CDS would be reduced to $40m
- Basket CDS would continue
- Premium paid each year:

22bp × $40m = $88,000 p.a.



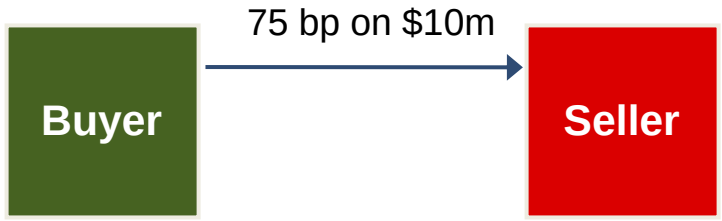


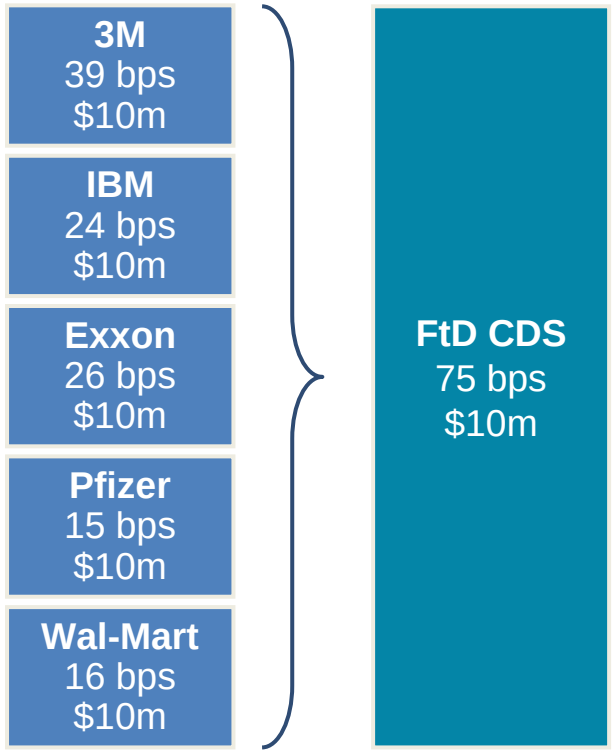
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First-to-Default CDS Example

- FtD CDS on notional portfolio of $50m
- Comprises 5 names of $10m notional each
- Premium for FtD less than sum of individual CDS premiums:
  - Total premium for 5 separate CDS: 120 bp
  - Premium for FtD CDS: 75 bp
- Premium paid each year:

  75bp × $10m = $75,000 p.a.



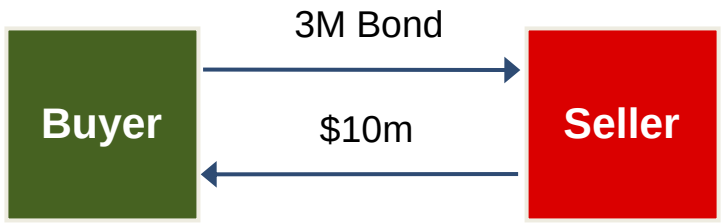


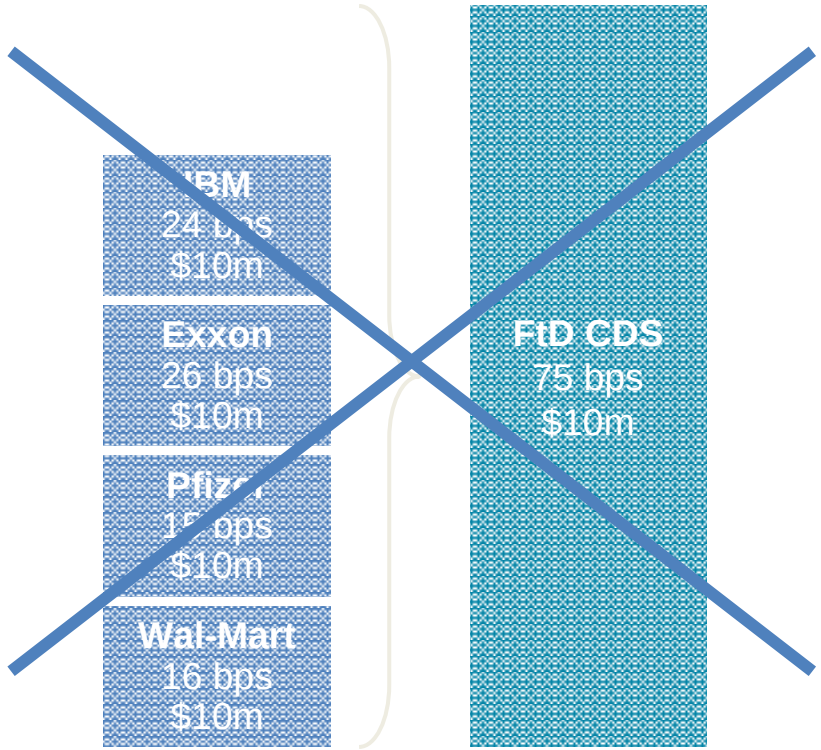
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First-to-Default CDS Example – After Default

Suppose 3M experiences a credit event…

- FtD CDS would make payment of $10m less recovery value
- FtD CDS then terminates
- Exposure of protection seller is limited to $10m, not $50m



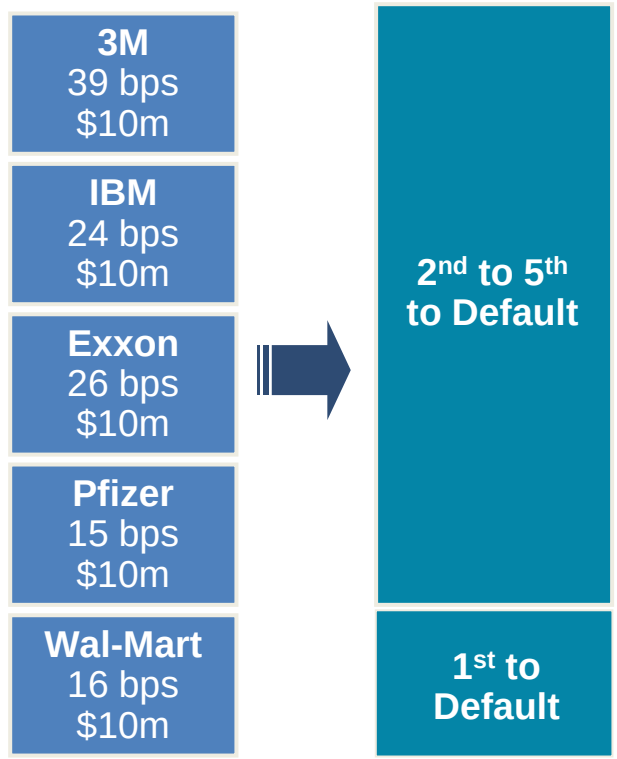


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Other Basket Swaps

- 2nd to default swap
  - Triggered by 2nd credit event
  - Terminates thereafter
- *N*th to default swap
  - Triggered by *n*th credit event
  - Terminates thereafter
- 2nd through default swap
  - Triggered by 2nd credit event
  - CDS continues to provide protection for subsequent defaults
  - Premium lowered pro rata after 2nd and subsequent defaults
- First *n* to default swap
  - Provides protection on first *n* defaults in basket
  - Terminates thereafter
- Last *p* to default swap
  - Provides protection on last *p* defaults in basket

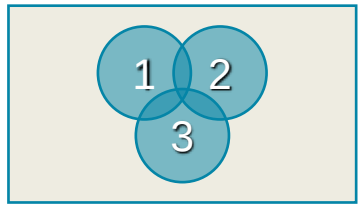
Illustration of 2nd through 5th to default



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FtD Swaps and Correlation

- Pricing depends on correlation
- Example: 3 reference assets each with PD = 10%
- If correlation = 0%
  - Probability of zero defaults P(0) = 0.93 = 72.9%
  - Probability of one or more defaults = 27.1%
    - P(1) = 24.3%
    - P(2) = 2.7%
    - P(3) = 0.1%
  - Probability of default is approximately equal to sum of default probabilities
  - **FtD premium approximately equal to sum of CDS premiums**



- If correlation = 100%
  - Probability of zero defaults P(0) = 0.9 = 90%
  - Probability of one or more defaults = 10%
  - Probability of default is that of entity with highest quoted CDS premium
  - **Equal to highest single-name CDS premium**

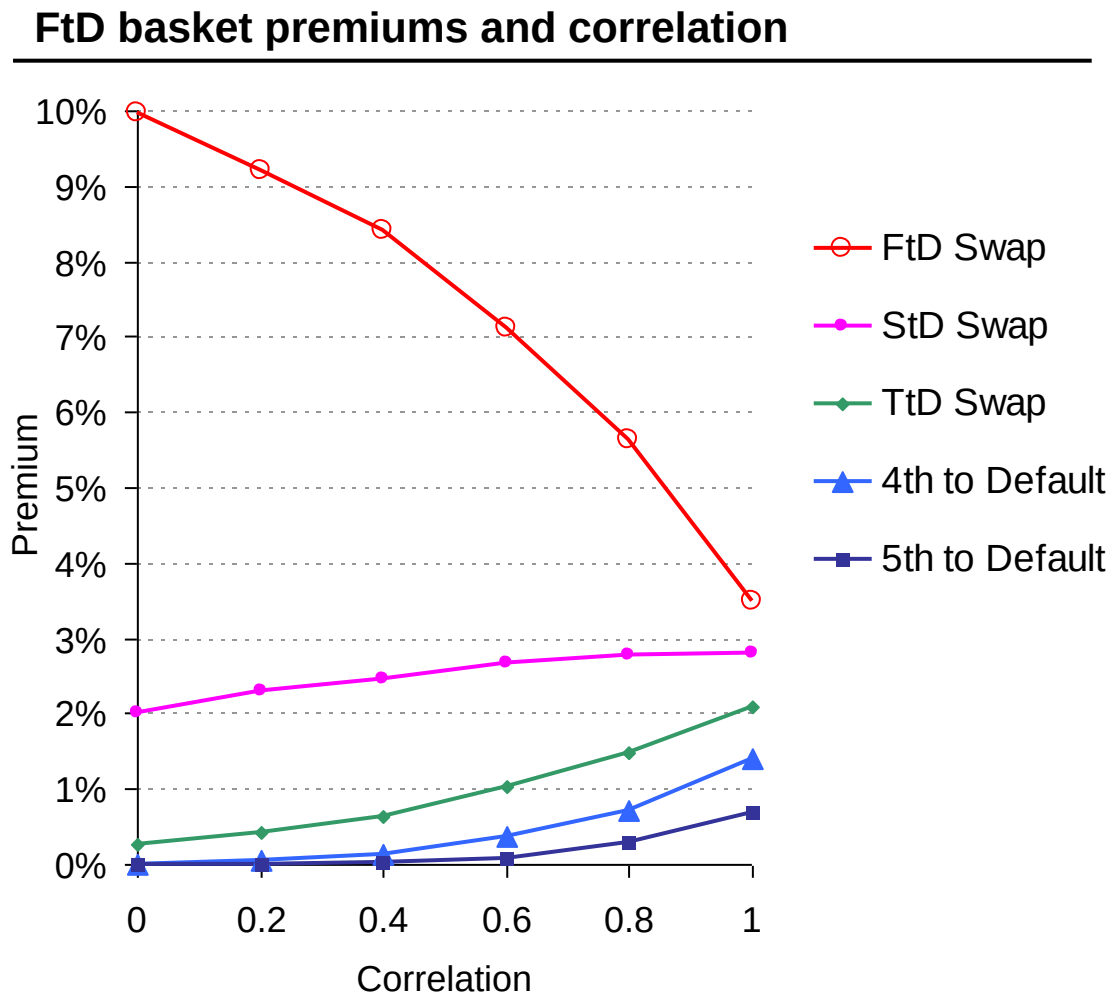


- For FtD swaps, as correlation goes up, probability of paying out goes down, so FtD swap will get cheaper

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FtD Swaps and Correlation

- Chart opposite shows CDS premiums for:
  - 1st to default swap
  - 2nd to default swap
  - 3rd to default swap
  - 4th to default swap
  - 5th to default swap
- … for correlation between 0 and 1
- Basket has 5 entities:
  - Bond 1 has a PD of 1%
  - Bond 2 has a PD of 2%
  - …
  - Bond 5 has a PD of 5%
  - Recovery Rate on all bonds is 30%
- If correlation=1, FtD premium is 350bp
- If correlation=0, FtD premium is 997bp



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